Data Quality Audit Report

superstore

Generated: March 16, 2026 at 22:55

Data Health Score

Metric	Value
Total Rows	9,994
Total Columns	21
Health Score	94.8/100 (Excellent)
Critical Issues (High)	2
Warnings (Medium)	2
Info Items (Low)	5
Columns with Missing Data	0
Duplicate Rows	0

Executive Summary

This report presents a comprehensive data quality audit of **superstore**, containing **9,994 rows** and **21 columns**. The overall data health score is **94.8/100** — rated **Excellent**.

A total of **2 high-severity**, **2 medium-severity**, and **5 low-severity** issues were identified across 5 quality-check categories.

Key findings: **Profit** — 1881 outliers (18.8%), valid range [-39.72, 70.82]; **Sales** — 1167 outliers (11.7%), valid range [-271.71, 498.93]; **Product Name** — Dominant casing covers only 54.1%: {'Title_Case': np.int64(5407), 'Mixed': np.int64(4545), 'ALL_CAPS': np.int64(36), 'all_lower': np.int64(6)}.

Immediate attention should be given to high-severity issues (missing values, duplicates, type inconsistencies) which have the greatest impact on analytical reliability. Detailed findings and prioritized recommendations follow.

Category	Weight	Score	Contribution
Completeness	35%	100.0	35.0
Uniqueness	15%	100.0	15.0
Type Consistency	20%	100.0	20.0
Outlier Reasonability	12%	93.2	11.2
Format Consistency	10%	75.0	7.5
Text Quality	8%	76.8	6.1
TOTAL	100%		94.8

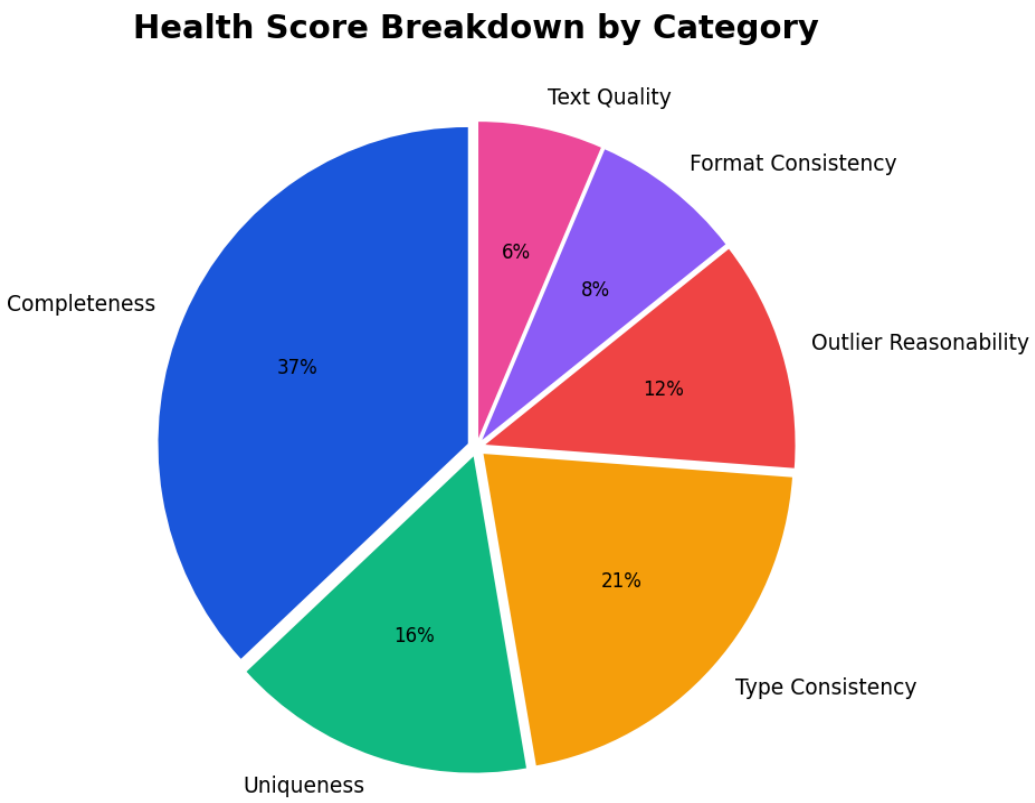

Figure 3: Health Score Breakdown by Quality Category

Findings Summary

The dataset **superstore** contains **9,994 rows** and **21 columns**, with an overall data health score of **94.8/100**, rated **Excellent**.

The audit identified **9 issues** in total — **2 high-severity**, **2 medium-severity**, and **5 low-severity** — spanning 5 quality-check categories.

Completeness scored **100.0/100**: no missing values were detected across any column.

Uniqueness scored **100.0/100**: no duplicate rows were detected.

Type Consistency scored a strong **100.0/100** with no type validation issues detected.

Format Consistency scored **75.0/100**: column **Order Date** contains 1 date format(s) detected: MM/DD/YYYY, which will cause parsing failures if not normalised to ISO 8601.

Text Quality scored **76.8/100**: inconsistent casing and/or leading/trailing whitespace were detected in **Product Name**.

1 column(s) have poor or auto-generated names (e.g. **Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Postal Code, Product ID, Sub-Category, Product Name**), which reduce maintainability and should be renamed or dropped.

Outlier Reasonability scored **93.2/100**: statistical outliers were found in 6 numeric column(s), with **Profit** being the most affected (18.8% of values outside IQR bounds).

The lowest-scoring quality dimension is **Format Consistency** at **75.0/100**, making it the primary focus area for remediation.

Addressing all 2 high-severity issues is estimated to raise the health score significantly; detailed remediation steps are provided in the Recommendations section.

Missing Values Analysis

No missing values detected.

Issue Severity Overview

The heatmap maps each quality check (rows) against each column (columns). **H** = High severity, **M** = Medium, **L** = Low, blank = no issue.

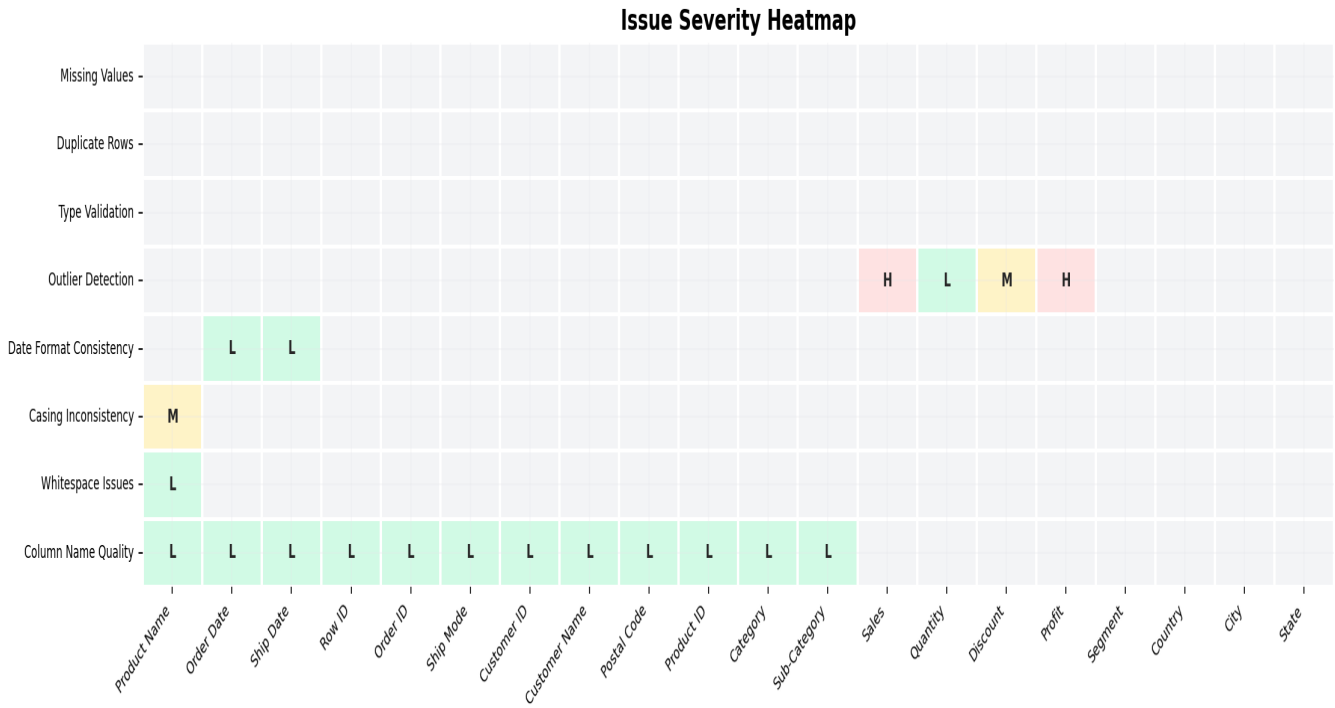

Figure 4: Issue Severity Heatmap

Top issue hotspots: **Product Name** (score 4), **Sales** (score 3), **Profit** (score 3), **Discount** (score 2), **Order Date** (score 2).

Distribution Analysis

Histograms for 6 numeric column(s). Red dashed = median; orange dotted = IQR outlier boundaries (Q1 – 1.5×IQR, Q3 + 1.5×IQR). Values beyond orange lines are statistical outliers.

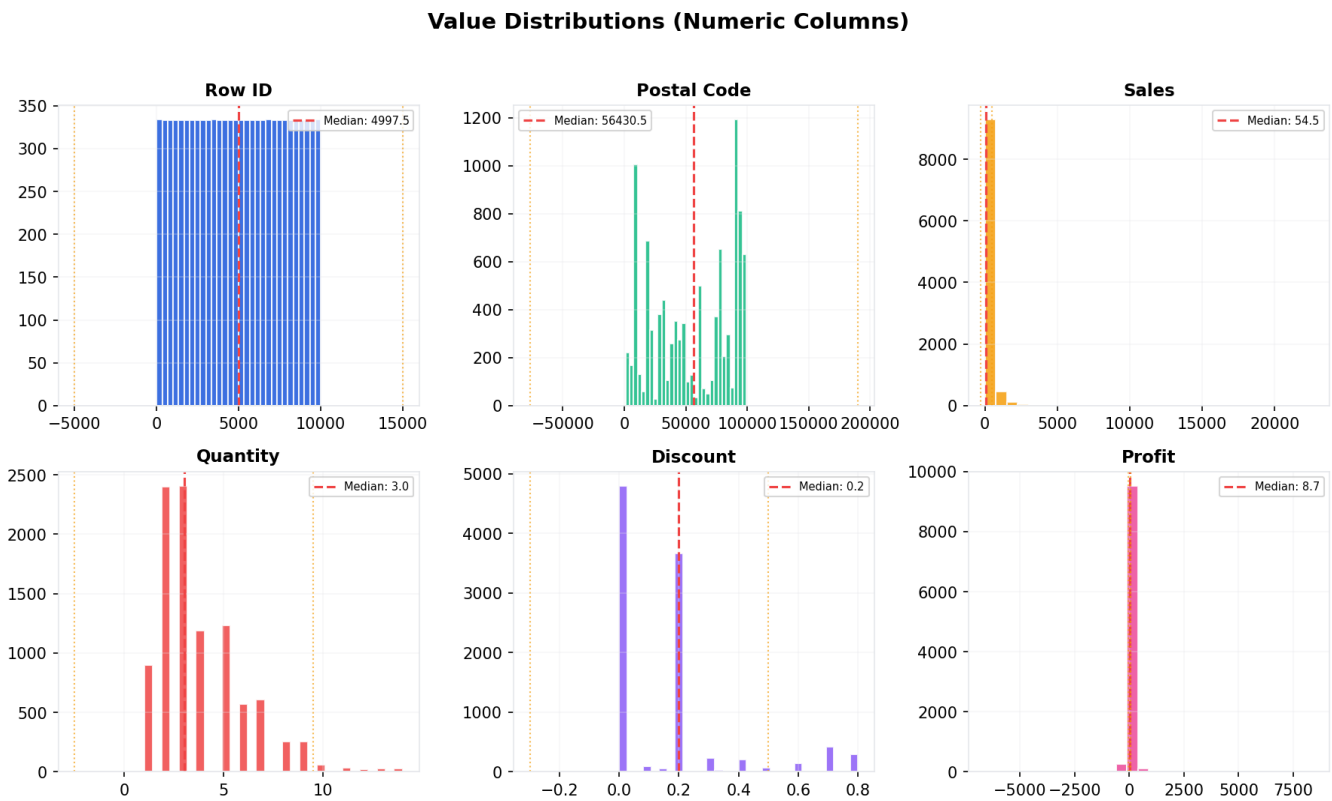

Figure 5: Value distributions for numeric columns

Sales: 1167 outlier(s) (11.7%), valid range [-271.71, 498.93].

Quantity: 170 outlier(s) (1.7%), valid range [-2.50, 9.50].

Discount: 856 outlier(s) (8.6%), valid range [-0.30, 0.50].

Profit: 1881 outlier(s) (18.8%), valid range [-39.72, 70.82].

Trend Analysis

Numeric values over time using **Order Date**. Raw data shown at low opacity; solid line = rolling average. Note: dates were parsed from mixed formats (MM/DD/YYYY, YYYY-MM-DD, DD-MM-YYYY).

Figure 6: Trends Over Time with rolling averages

Detailed Findings

All 9 issues sorted by severity (High → Low), then affected rows.

Column	Check Type	Severity	Finding	Rows	%
Profit	Outlier Detection	High	1881 outliers (18.8%), valid range [-39.72, 70.82]	1881	18.8%
Sales	Outlier Detection	High	1167 outliers (11.7%), valid range [-271.71, 498.93]	1167	11.7%
Product Name	Casing Inconsistency	Medium	Dominant casing covers only 54.1%: {'Title_Case': np.5	4587	45.9%
Discount	Outlier Detection	Medium	856 outliers (8.6%), valid range [-0.30, 0.50]	856	8.6%
Order Date	Date Format Consistenc	Low	1 date format(s) detected: MM/DD/YYYY	9994	100.0%
Ship Date	Date Format Consistenc	Low	1 date format(s) detected: MM/DD/YYYY	9994	100.0%
Row ID, Order ID, Orde	Column Name Quality	Low	11 column(s) with spaces/special chars in name	9994	100.0%
Quantity	Outlier Detection	Low	170 outliers (1.7%), valid range [-2.50, 9.50]	170	1.7%
Product Name	Whitespace Issues	Low	48 values with leading/trailing or double spaces (0.5%)	48	0.5%

Recommendations

Priority	Issue	Recommendation	Affected Columns	Est. Impact
P2 — High	Inconsistent text casing	Apply .str.title() to name/category columns. Pick one convention and enforce it in the ingestion pipeline.	Product Name	+5 pts on Text Quality
P3 — Medium	Outliers in numeric columns (worst: Profit)	Investigate outlier rows — validate against source data. Use IQR capping (Winsorization) for extreme values. Do NOT blindly drop — confirm business context first.	Row ID, Postal Code, Sales	+1.0 pts on Outlier Reasonability